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Surface-acoustic-wave (SAW) filter with dielectric material disposed in a piezoelectric layer

Abstract

An apparatus is disclosed for implementing a surface-acoustic-wave (SAW) filter with dielectric material disposed in a piezoelectric layer. In an example aspect, the apparatus includes a surface-acoustic-wave filter with a piezoelectric layer, an electrode structure, and dielectric material. The piezoelectric layer has at least one channel. The dielectric material is disposed in the at least one channel of the piezoelectric layer and is at least partially covered by the electrode structure.

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Background/Summary

TECHNICAL FIELD

(1) This disclosure relates generally to wireless transceivers and other components that employ filters and, more specifically, to a surface-acoustic-wave (SAW) filter with dielectric material disposed in at least one channel of a piezoelectric layer.

BACKGROUND

(2) Electronic devices use radio-frequency (RF) signals to communicate information. These radio-

frequency signals enable users to talk with friends, download information, share pictures, remotely control household devices, and receive global positioning information. To transmit or receive the radio-frequency signals within a given frequency band, the electronic device may use filters to pass signals within the frequency band and to suppress (e.g., attenuate) jammers or noise having frequencies outside of the frequency band. It can be challenging, however, to design a filter that provides filtering for radio-frequency applications, including those that utilize frequencies above 2 gigahertz (GHz).

SUMMARY

(3) An apparatus is disclosed that implements a surface-acoustic-wave (SAW) filter with dielectric material disposed in a piezoelectric layer. Instead of increasing the mass of an electrode structure within a trap region of a surface-acoustic-wave filter, example techniques use a piezoelectric layer with at least one channel present within the trap region. Dielectric material is disposed in the channel and enables the surface-acoustic-wave filter to operate in a piston mode. With the dielectric material at least partially embedded within the piezoelectric layer, the surface-acoustic-wave filter can attenuate spurious modes across a wider range of frequencies compared to other spurious-mode suppression techniques. Additionally, disposing the dielectric material within the channel of the piezoelectric layer can be easier to manufacture, especially for high-frequency applications, compared to other spurious-mode suppression techniques.

(4) In an example aspect, an apparatus for filtering is disclosed. The apparatus includes a surface-acoustic-wave filter with a piezoelectric layer, an electrode structure, and dielectric material. The piezoelectric layer has at least one channel. The dielectric material is disposed in the at least one channel of the piezoelectric layer and is at least partially covered by the electrode structure.

(5) In an example aspect, an apparatus for filtering is disclosed. The apparatus includes a surface-acoustic-wave filter configured to generate a filtered signal from a radio-frequency signal. The surface-acoustic-wave filter includes means for converting the radio-frequency signal to an acoustic wave and converting a propagated acoustic wave into the filtered signal. The surface-acoustic-wave filter also includes means for propagating the acoustic wave across a planar surface to produce the propagated acoustic wave. The surface-acoustic-wave filter additionally includes means for operating in accordance with a piston mode. The means for operating is disposed in the means for propagating and is at least partially covered by the means for converting.

(6) In an example aspect, a method for manufacturing a surface-acoustic-wave filter is disclosed. The method includes providing a piezoelectric layer. The method also includes etching at least one channel across a surface of the piezoelectric layer. Additionally, the method includes depositing dielectric material within the at least one channel. The method further includes providing an electrode structure that at least partially covers the dielectric material.

(7) In an example aspect, an apparatus is disclosed. The apparatus include a piezoelectric layer and dielectric material. The piezoelectric layer is configured to propagate an acoustic wave along a first axis. The piezoelectric layer has at least one recessed area with a longitudinal axis that is substantially parallel to the first axis. The dielectric material is positioned in the recessed area and has at least one surface exposed.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. **1** illustrates an example operating environment for a surface-acoustic-wave filter with dielectric material disposed in a piezoelectric layer.

(2) FIG. **2** illustrates an example wireless transceiver including at least one surface-acoustic-wave filter with dielectric material disposed in a piezoelectric layer.

(3) FIG. **3** illustrates example components of a surface-acoustic-wave filter with dielectric material

disposed in a piezoelectric layer.
(4) FIG. **4-1** illustrates an example implementation of a thin-film surface-acoustic-wave filter with dielectric material disposed in a piezoelectric layer.
(5) FIG. **4-2** illustrates an example implementation of a high-quality temperature-compensated surface-acoustic-wave filter with dielectric material disposed in a piezoelectric layer.
(6) FIG. **5** illustrates an example electrode structure of a surface-acoustic-wave filter with dielectric material disposed in a piezoelectric layer.
(7) FIG. **6** illustrates a first example implementation of a surface-acoustic-wave filter with dielectric material disposed in a piezoelectric layer.
(8) FIG. **7** illustrates a second example implementation of a surface-acoustic-wave filter with dielectric material disposed in a piezoelectric layer.
(9) FIG. **8** illustrates a third example implementation of a surface-acoustic-wave filter with dielectric material disposed in a piezoelectric layer.
(10) FIG. **9** illustrates a fourth example implementation of a surface-acoustic-wave filter with dielectric material disposed in a piezoelectric layer.
(11) FIG. **10** illustrates example frequency responses for implementations of a surface-acoustic-wave filter that does and does not include dielectric material disposed in a piezoelectric layer.
(12) FIG. **11** is a flow diagram illustrating an example process for manufacturing a surface-acoustic-wave filter having dielectric material disposed in a piezoelectric layer.

DETAILED DESCRIPTION

(13) To transmit or receive radio-frequency signals within a given frequency band, an electronic device may use filters to pass signals within the frequency band and to suppress (e.g., attenuate) jammers or noise having frequencies outside of the frequency band. Electroacoustic devices (e.g., "acoustic filters") can be used to filter high-frequency signals in many applications, such as those with frequencies that are greater than 100 megahertz (MHz). An acoustic filter is tuned to pass certain frequencies (e.g., frequencies within its passband) and attenuate other frequencies (e.g., frequencies that are outside of its passband). Using piezoelectric material as a vibrating medium, the acoustic filter operates by transforming an electrical signal wave that is propagating along an electrical conductor into an acoustic wave (e.g., an acoustic signal wave) that forms across the piezoelectric material. The acoustic wave is then converted back into an electrical filtered signal. The acoustic filter can include an electrode structure that transforms or converts between the electrical and acoustic waves.
(14) The acoustic wave propagates across the piezoelectric material at a velocity having a magnitude that is significantly less than that of the propagation velocity of the electromagnetic wave. Generally, the magnitude of the propagation velocity of a wave is proportional to a size of a wavelength of the wave. Consequently, after conversion of the electrical signal wave into the acoustic signal wave, the wavelength of the acoustic signal wave is significantly smaller than the wavelength of the electrical signal wave. The resulting smaller wavelength of the acoustic signal wave enables filtering to be performed using a smaller filter device. This permits acoustic filters to be used in space-constrained devices, including portable electronic devices such as cellular phones.
(15) It can be challenging, however, to design an acoustic filter that can provide filtering for high-frequency applications, including those that utilize frequencies above 2 gigahertz (GHz), while maintaining or reducing spurious modes (e.g., spurious wave modes) below certain levels within a passband. A spurious mode is an undesired mode, which can degrade performance of the acoustic filter. Some filter designs customize a geometry of the electrode structure of the acoustic filter to attenuate spurious modes. As an example, fingers within the electrode structure can have varying widths or heights across the length of the fingers. In particular, a finger can have a wider and/or taller profile (e.g., include a hammerhead and dot, respectively) within a trap region of the acoustic filter compared to an active track region of the acoustic filter. Within the trap region and the active track region, the projection of adjacent fingers of the acoustic filter overlap. The trap region

borders the active track region and can include ends of the fingers. The addition of hammerheads and dots to the fingers increase the mass of the electrode structure within the trap region relative to the active track region. Customizing the electrode structure in this way can attenuate (e.g., suppress) some types of spurious modes.

(16) Although this technique enables suppression of a spurious mode, it can be challenging to apply this technique to acoustic filters that support higher frequencies, such as those used in frequency bands 7, 40, 41, n77, and n79; with Wi Fi® at 2.4 GHz; with 5 GHz frequencies; and/or at sub-6 GHz frequencies. In particular, a size of the hammerhead or dot within the trap region can become too small or potentially less practical or costly for the manufacturing equipment in some scenarios to accurately be produced due to lithographic constraints.

(17) To overcome these difficulties, another technique applies a dielectric bar across fingers of the electrode structure to suppress spurious modes. In some implementations, the dielectric bar has a thickness on the order of tens of nanometers, such as approximately 20 nanometers. The dielectric bar, however, can be affected by subsequent manufacturing processes, such as trimming or passivation. Also, the dielectric bar may be unable to provide sufficient spurious-mode suppression across a desired range of frequencies.

(18) To address these challenges, example techniques for implementing a surface-acoustic-wave (SAW) filter with dielectric material disposed in a piezoelectric layer are described. Instead of or in addition to increasing the mass of an electrode structure within a trap region of a surface-acoustic-wave filter, example techniques use a piezoelectric layer with at least one channel present within the trap region. Dielectric material is disposed in the channel and enables the surface-acoustic-wave filter to operate in a piston mode.

(19) There are several benefits that result from certain implementations of a surface-acoustic-wave filter with dielectric material disposed in the piezoelectric layer. These benefits can include operational improvements, manufacturing advantages, or both. One operational benefit is the suppression of spurious modes. In particular, the dielectric material causes an acoustic wave to have a lower velocity within the trap region in comparison to the active track region. The lower velocity within the trap region suppresses spurious modes, such as a transversal mode. By tailoring the geometry of the dielectric material within the trap region, the dielectric material can attenuate spurious modes across a wider range of frequencies compared to other techniques.

(20) From a manufacturing perspective, the techniques for disposing dielectric material within the piezoelectric layer can be potentially easier to implement compared to the techniques for applying hammerheads or dots, especially for high-frequency applications. Also, in contrast to the dielectric bars, embedding the dielectric material within the piezoelectric layer creates the acoustic-wave confinement prior to the realization of the metallization of the electrode structure. As the electrode structure at least partially covers the dielectric material, the dielectric material is less exposed to the subsequent manufacturing processes, such as trimming and passivation, compared to the dielectric bar. In this way, the surface-acoustic-wave filter with the dielectric material disposed in the piezoelectric layer can experience less performance degradation to the piston mode by the subsequent processing steps compared to another surface-acoustic-wave filter with the dielectric bar. Also, the surface-acoustic-wave filter with the dielectric material disposed in the piezoelectric layer can experience less electrical losses due to less subsequent fabrication steps after deposition of the electrode structure compared to the other surface-acoustic-wave filter with the dielectric bar.

(21) FIG. **1** illustrates an example environment **100** for operating a surface-acoustic-wave filter with dielectric material disposed in a piezoelectric layer. In the environment **100**, a computing device **102** communicates with a base station **104** through a wireless communication link **106** (wireless link **106**). In this example, the computing device **102** is depicted as a smartphone. However, the computing device **102** can be implemented as any suitable computing or electronic device, such as a modem, a cellular base station, a broadband router, an access point, a cellular phone, a gaming device, a navigation device, a media device, a laptop computer, a desktop

computer, a tablet computer, a wearable computer, a server, a network-attached storage (NAS) device, a smart appliance or other internet of things (IoT) device, a medical device, a vehicle-based communication system, a radar, a radio apparatus, and so forth.

(22) The base station **104** communicates with the computing device **102** via the wireless link **106**, which can be implemented as any suitable type of wireless link. Although depicted as a tower of a cellular network, the base station **104** can represent or be implemented as another device, such as a satellite, a server device, a terrestrial television broadcast tower, an access point, a peer-to-peer device, a mesh network node, and so forth. Therefore, the computing device **102** may communicate with the base station **104** or another device via a wireless connection.

(23) The wireless link **106** can include a downlink of data or control information communicated from the base station **104** to the computing device **102**, an uplink of other data or control information communicated from the computing device **102** to the base station **104**, or both a downlink and an uplink. The wireless link **106** can be implemented using any suitable communication protocol or standard, such as 2nd-generation (2G), 3rd-generation (3G), 4th-generation (4G), or 5th-generation (5G) cellular; IEEE 802.11 (e.g., Wi-Fi®); IEEE 802.15 (e.g., Bluetooth®); IEEE 802.16 (e.g., WiMAX®); and so forth. In some implementations, the wireless link **106** may wirelessly provide power and the base station **104** or the computing device **102** may comprise a power source.

(24) As shown, the computing device **102** includes an application processor **108** and a computer-readable storage medium **110** (CRM **110**). The application processor **108** can include any type of processor, such as a multi-core processor, that executes processor-executable code stored by the CRM **110**. The CRM **110** can include any suitable type of data storage media, such as volatile memory (e.g., random access memory (RAM)), non-volatile memory (e.g., Flash memory), optical media, magnetic media (e.g., disk), and so forth. In the context of this disclosure, the CRM **110** is implemented to store instructions **112**, data **114**, and other information of the computing device **102**, and thus does not include transitory propagating signals or carrier waves.

(25) The computing device **102** can also include input/output ports **116** (I/O ports **116**) and a display **118**. The I/O ports **116** enable data exchanges or interaction with other devices, networks, or users. The I/O ports **116** can include serial ports (e.g., universal serial bus (USB) ports), parallel ports, audio ports, infrared (IR) ports, user interface ports such as a touchscreen, and so forth. The display **118** presents graphics of the computing device **102**, such as a user interface associated with an operating system, program, or application. Alternatively or additionally, the display **118** can be implemented as a display port or virtual interface, through which graphical content of the computing device **102** is presented.

(26) A wireless transceiver **120** of the computing device **102** provides connectivity to respective networks and other electronic devices connected therewith. The wireless transceiver **120** can facilitate communication over any suitable type of wireless network, such as a wireless local area network (WLAN), peer-to-peer (P2P) network, mesh network, cellular network, ultra-wideband (UWB) network, wireless wide-area-network (WWAN), and/or wireless personal-area-network (WPAN). In the context of the example environment **100**, the wireless transceiver **120** enables the computing device **102** to communicate with the base station **104** and networks connected therewith. However, the wireless transceiver **120** can also enable the computing device **102** to communicate “directly” with other devices or networks.

(27) The wireless transceiver **120** includes circuitry and logic for transmitting and receiving communication signals via an antenna **122**. Components of the wireless transceiver **120** can include amplifiers, switches, mixers, analog-to-digital converters, filters, and so forth for conditioning the communication signals (e.g., for generating or processing signals). The wireless transceiver **120** can also include logic to perform in-phase/quadrature (I/Q) operations, such as synthesis, encoding, modulation, decoding, demodulation, and so forth. In some cases, components of the wireless transceiver **120** are implemented as separate transmitter and receiver entities. Additionally or

alternatively, the wireless transceiver **120** can be realized using multiple or different sections to implement respective transmitting and receiving operations (e.g., separate transmit and receive chains). In general, the wireless transceiver **120** processes data and/or signals associated with communicating data of the computing device **102** over the antenna **122**.
(28) In the example shown in FIG. **1**, the wireless transceiver **120** includes at least one surface-acoustic-wave filter **124** (SAW filter **124**). In some implementations, the wireless transceiver **120** includes multiple surface-acoustic-wave filters **124**, which can be formed from surface-acoustic-wave resonators arranged in series, in parallel, in a ladder structure, in a lattice structure, or some combination thereof. The surface-acoustic-wave filter **124** can be a thin-film surface-acoustic-wave filter **126** (TFSAW filter **126**), a high-quality temperature-compensated surface-acoustic-wave filter (HQ-TC SAW filter), or another type of surface-acoustic-wave filter not shown.
(29) The surface-acoustic-wave filter **124** includes at least one electrode structure **130**, at least one piezoelectric layer **132** (or piezoelectric material), and dielectric material **134** (e.g., a dielectric or at least one dielectric layer). The dielectric material **134** is at least partially embedded within the piezoelectric layer **132** and is at least partially covered by the electrode structure **130**. With this dielectric material **134**, the surface-acoustic-wave filter **124** can operate in a piston mode and attenuate spurious modes, such as a transversal mode. In general, the piston mode is characterized by a longitudinal velocity that varies in a transverse direction (e.g., along a second (Y) axis **412** described with respect to FIGS. **4-1** to **9**). The piston mode reduces energy losses by reducing acoustic-wave propagation outside an active track region of the surface-acoustic-wave filter **124** in the transverse direction and by reducing transversally standing waves within the active track region. The surface-acoustic-wave filter **124** is further described with respect to FIG. **2**.
(30) FIG. **2** illustrates an example wireless transceiver **120**. In the depicted configuration, the wireless transceiver **120** includes a transmitter **202** and a receiver **204**, which are respectively coupled to a first antenna **122-1** and a second antenna **122-2**. In other implementations, the transmitter **202** and the receiver **204** can be connected to a same antenna through a duplexer (not shown). The transmitter **202** is shown to include at least one digital-to-analog converter **206** (DAC **206**), at least one first mixer **208-1**, at least one amplifier **210** (e.g., a power amplifier), and at least one first surface-acoustic-wave filter **124-1**. The receiver **204** includes at least one second surface-acoustic-wave filter **124-2**, at least one amplifier **212** (e.g., a low-noise amplifier), at least one second mixer **208-2**, and at least one analog-to-digital converter **214** (ADC **214**). The first mixer **208-1** and the second mixer **208-2** are coupled to a local oscillator **216**. Although not explicitly shown, the digital-to-analog converter **206** of the transmitter **202** and the analog-to-digital converter **214** of the receiver **204** can be coupled to the application processor **108** (of FIG. **1**) or another processor associated with the wireless transceiver **120** (e.g., a modem).
(31) In some implementations, the wireless transceiver **120** is implemented using multiple circuits (e.g., multiple integrated circuits), such as a transceiver circuit **236** and a radio-frequency front-end (RFFE) circuit **238**. As such, the components that form the transmitter **202** and the receiver **204** are distributed across these circuits. As shown in FIG. **2**, the transceiver circuit **236** includes the digital-to-analog converter **206** of the transmitter **202**, the mixer **208-1** of the transmitter **202**, the mixer **208-2** of the receiver **204**, and the analog-to-digital converter **214** of the receiver **204**. In other implementations, the digital-to-analog converter **206** and the analog-to-digital converter **214** can be implemented on another separate circuit that includes the application processor **108** or the modem. The radio-frequency front-end circuit **238** includes the amplifier **210** of the transmitter **202**, the surface-acoustic-wave filter **124-1** of the transmitter **202**, the surface-acoustic-wave filter **124-2** of the receiver **204**, and the amplifier **212** of the receiver **204**.
(32) During transmission, the transmitter **202** generates a radio-frequency transmit signal **218**, which is transmitted using the antenna **122-1**. To generate the radio-frequency transmit signal **218**, the digital-to-analog converter **206** provides a pre-upconversion transmit signal **220** to the first mixer **208-1**. The pre-upconversion transmit signal **220** can be a baseband signal or an

intermediate-frequency signal. The first mixer **208-1** upconverts the pre-upconversion transmit signal **220** using a local oscillator (LO) signal **222** provided by the local oscillator **216**. The first mixer **208-1** generates an upconverted signal, which is referred to as a pre-filter transmit signal **224**. The pre-filter transmit signal **224** can be a radio-frequency signal and include some noise or unwanted frequencies, such as a harmonic frequency. The amplifier **210** amplifies the pre-filter transmit signal **224** and passes the amplified pre-filter transmit signal **224** to the first surface-acoustic-wave filter **124-1**.

(33) The first surface-acoustic-wave filter **124-1** filters the amplified pre-filter transmit signal **224** to generate a filtered transmit signal **226**. As part of the filtering process, the first surface-acoustic-wave filter **124-1** attenuates the noise or unwanted frequencies within the pre-filter transmit signal **224**. The transmitter **202** provides the filtered transmit signal **226** to the antenna **122-1** for transmission. The transmitted filtered transmit signal **226** is represented by the radio-frequency transmit signal **218**.

(34) During reception, the antenna **122-2** receives a radio-frequency receive signal **228** and passes the radio-frequency receive signal **228** to the receiver **204**. The second surface-acoustic-wave filter **124-2** accepts the received radio-frequency receive signal **228**, which is represented by a pre-filter receive signal **230**. The second surface-acoustic-wave filter **124-2** filters any noise or unwanted frequencies within the pre-filter receive signal **230** to generate a filtered receive signal **232**.

(35) The amplifier **212** of the receiver **204** amplifies the filtered receive signal **232** and passes the amplified filtered receive signal **232** to the second mixer **208-2**. The second mixer **208-2** downconverts the amplified filtered receive signal **232** using the local oscillator signal **222** to generate the downconverted receive signal **234**. The analog-to-digital converter **214** converts the downconverted receive signal **234** into a digital signal, which can be processed by the application processor **108** or another processor associated with the wireless transceiver **120** (e.g., the modem).

(36) FIG. **2** illustrates one example configuration of the wireless transceiver **120**. Other configurations of the wireless transceiver **120** can support multiple frequency bands and share an antenna **122** across multiple transceivers. One of ordinary skill in the art can appreciate the variety of other configurations for which surface-acoustic-wave filters **124** may be included. For example, the surface-acoustic-wave filters **124** can be integrated within duplexers or diplexers of the wireless transceiver **120**. Example implementations of the surface-acoustic-wave filter **124-1** or **124-2** are further described with respect to FIGS. **3** to **9**.

(37) FIG. **3** illustrates example components of the surface-acoustic-wave filter **124**. In the depicted configuration, the surface-acoustic-wave filter **124** includes the electrode structure **130**, the piezoelectric layer **132**, the dielectric material **134**, and at least one substrate layer **302**. The electrode structure **130** comprises an electrically conductive material, such as metal, and can include one or more layers. The one or more layers can include one or more metal layers and can optionally include one or more adhesion layers. As an example, the metal layers can be composed of aluminium (Al), copper (Cu), silver (Ag), gold (Au), tungsten (W), platinum (Pt), or some combination or doped version thereof. The adhesion layers can be composed of chromium (Cr), titanium (Ti), molybdenum (Mo), or some combination thereof.

(38) The electrode structure **130** can include one or more interdigital transducers **304**. The interdigital transducer **304** converts an electrical signal into an acoustic wave and converts the acoustic wave into a filtered electrical signal. The interdigital transducer **304** includes at least two comb-shaped structures **306-1** and **306-2**. Each comb-shaped structure **306-1** and **306-2** includes a busbar **308** (e.g., a conductive segment or rail) and multiple fingers **310** (e.g., electrode fingers). An example interdigital transducer **304** is further described with respect to FIGS. **4-1** and **4-2**. Although not explicitly shown, the electrode structure **130** can also include two or more reflectors. In an example implementation, the interdigital transducer **304** is arranged between two reflectors, which reflect the acoustic wave back towards the interdigital transducer **304**.

(39) In example implementations, the piezoelectric layer **132** can be implemented using a variety of

different materials that exhibit piezoelectric properties (e.g., can transfer mechanical energy into electrical energy or electrical energy into mechanical energy). Example types of material include lithium niobate (LiNbO.sub.3), lithium tantalate (LiTaO.sub.3), quartz, aluminium nitride (AlN), aluminium scandium nitride (AlScN), or some combination thereof. In general, the material that forms the piezoelectric layer **132** has a crystalline structure. This crystalline structure is defined by an ordered arrangement of particles (e.g., atoms, ions, or molecules).

(40) The piezoelectric layer **132** has at least one channel **312** (e.g., at least one recess area or at least one trench) on a surface that faces the electrode structure **130**. The channel **312** can be proximate to the beginning and/or ends of the fingers **310** of the electrode structure **130**. The term “proximate” can refer to the channel **312** being closer to the beginning or ends of the fingers **310** compared to a middle or center portion of the fingers **310**. Explained another way, the channel **312** is aligned with the beginning and/or ends of the finger **310** such that it is disposed within a trap region of the surface-acoustic-wave filter **124**, which is further described with respect to FIG. **5**. In general, the channel **312** can pass “below” one or more fingers **310** of the electrode structure **130**.

(41) In some implementations, the channel **312** includes one channel that is proximate to a boundary of an active track region of the surface-acoustic-wave filter **124**, as further described with respect to FIG. **6**. In other implementations, the channel **312** is implemented using two or more channels that are substantially parallel and are proximate to a boundary of the active track region, as further described with respect to FIG. **7**. In still other implementations, the channel **312** includes multiple segmented channels that are aligned along an axis that is substantially parallel to the direction of acoustic-wave propagation and are proximate to a boundary of the active track region. These segmented channels can be partially covered by the electrode structure **130**, as described with respect to FIG. **8**, fully covered by the electrode structure **130**, as described with respect to FIG. **9**, or some combination thereof. In some implementations, a longitudinal axis of the channel **312** is substantially orthogonal to longitudinal axes of the fingers **310**. Also, the longitudinal axis of the channel **312** can be substantially parallel to the direction of acoustic-wave propagation, as further described with respect to FIGS. **4-1** and **4-2**.

(42) The dielectric material **134** can be formed using a variety of different types of dielectric material, such as tantalum oxide (Ta.sub.2O.sub.5), silicon oxide (SiO), silicon dioxide (SiO.sub.2), silicon nitride (SiN), hafnium oxide (HfO.sub.2), aluminum oxide (Al.sub.2O.sub.3), tungsten trioxide (WO.sub.3), niobium pentoxide (Nb.sub.2O.sub.5), or some combination or doped version thereof. In some implementations, the dielectric material **134** is implemented using multiple layers formed using the same material or different materials. The dielectric material **134** can be at least partially buried or embedded within the piezoelectric layer **132**. In particular, the dielectric material **134** is disposed in (or positioned in) the one or more channels **312** of the piezoelectric layer **132**. In general, a surface of the dielectric material **134** that faces the electrode structure **130** is exposed or not covered by the piezoelectric layer **132**. In some implementations, at least a portion of the surface of the dielectric material **134** can be in direct contact with the electrode structure **130**. In alternative implementations, a dielectric layer (not shown) can be present between the electrode structure **130** and the dielectric material **134**. In example implementations, a thickness of this dielectric layer is approximately 10 nanometers or less (e.g., 10, 8, 5, or 2 nanometers). In general, the dielectric material **134** is at least partially covered by the electrode structure **130**.

(43) The substrate layer **302** includes one or more sublayers that can support passivation, temperature compensation, power handling, mode suppression, and so forth. As an example, the substrate layer **302** can include at least one compensation layer **314**, at least one charge-trapping layer **316**, at least one support layer **318**, or some combination thereof. These sublayers can be considered part of the substrate layer **302** or their own separate layers.

(44) The compensation layer **314** can provide temperature compensation to enable the surface-acoustic-wave filter **124** to achieve a target temperature coefficient of frequency based on the

thickness of the piezoelectric layer **132**. In some implementations, a thickness of the compensation layer **314** can be tailored to provide mode suppression (e.g., suppress the spurious plate mode). In example implementations, the compensation layer **314** can be implemented using at least one silicon dioxide (SiO2) layer, at least one doped silicon dioxide layer, at least one silicon nitride layer, at least one silicon oxynitride layer, or some combination thereof. In some applications, the substrate layer **302** may not include, for instance, the compensation layer **314** to reduce cost of the surface-acoustic-wave filter **124**.
(45) The charge-trapping layer **316** can trap induced charges at the interface between the compensation and support layer in order to, for example, suppress nonlinear substrate effects. The charge-trapping layer **316** can include at least one polysilicon (poly-Si) layer (e.g., a polycrystalline silicon layer or a multicrystalline silicon layer), at least one amorphous silicon layer, at least one silicon nitride (SiN) layer, at least one silicon oxynitride (SiON) layer, at least one aluminium nitride (AlN) layer, diamond-like carbon (DLC), diamond, or some combination thereof.
(46) The support layer **318** can enable the acoustic wave to form across the surface of the piezoelectric layer **132** and reduce the amount of energy that leaks into the substrate layer **302**. In some implementations, the support layer **318** can also act as a compensation layer **314**. In general, the support layer **318** is composed of material that is non-conducting and provides isolation. For example, the support layer **318** can be formed using silicon (Si) material (e.g., a doped high-resistive silicon material), sapphire material (e.g., aluminium oxide (Al.sub.2O.sub.3)), silicon carbide (SiC) material, fused silica material, quartz, glass, diamond, or some combination thereof. In some implementations, the support layer **318** has a relatively similar thermal expansion coefficient (TEC) as the piezoelectric layer **132**. The support layer **318** can also have a particular crystal orientation to support the suppression or attenuation of spurious modes.
(47) In some aspects, the surface-acoustic-wave filter **124** can be considered a resonator. Sometimes the surface-acoustic-wave filter **124** can be connected to other resonators associated with different layer stacks than the surface-acoustic-wave filter **124**. In other aspects, the surface-acoustic-wave filter **124** can be implemented as multiple interconnected resonators, which use the same layers (e.g., the piezoelectric layer **132** and/or the substrate layer **302**). The electrode structure **130**, the piezoelectric layer **132**, and the dielectric material **134** are further described with respect to FIGS. **4-1** and **4-2**.
(48) FIG. **4-1** illustrates an example implementation of the thin-film surface-acoustic-wave filter **126** with the dielectric material **134** disposed in the piezoelectric layer **132**. A three-dimensional perspective view **400-1** of the thin-film surface-acoustic-wave filter **126** is shown at the top of FIG. **4-1**, and a two-dimensional cross-section view **400-2** of the thin-film surface-acoustic-wave filter **126** is shown at the bottom of FIG. **4-1**.
(49) The thin-film surface-acoustic-wave filter **126** includes at least one electrode structure **130**, at least one piezoelectric layer **132**, and at least one substrate layer **302**. The electrode structure **130** can include one or more interdigital transducers **304**. In the depicted configuration shown in the two-dimensional cross-section view **400-2**, the piezoelectric layer **132** is disposed between the electrode structure **130** and the substrate layer **302**.
(50) In the three-dimensional perspective view **400-1**, the interdigital transducer **304** is shown to have the two comb-shaped structures **306-1** and **306-2** with fingers **310** extending from two busbars **308** towards each other. The fingers **310** are arranged in an interlocking manner in between the two busbars **308** of the interdigital transducer **304** (e.g., arranged in an interdigitated manner). In other words, the fingers **310** connected to a first busbar **308** extend towards a second busbar **308** but do not connect to the second busbar **308**. As such, there is a barrier region **402** (e.g., a transversal gap region) between the ends of these fingers and the second busbar **308**. Likewise, fingers **310** connected to the second busbar **308** extend towards the first busbar **308** but do not connect to the first busbar **308**. There is therefore a barrier region **402** between the ends of these fingers **310** and the first busbar **308**, as further described with respect to FIG. **5**.

(51) In the direction along the busbars **308**, there is an overlap region including a central region **404** where a portion of one finger **310** overlaps with a portion of an adjacent finger **310**. This central region **404**, including the overlap, may be referred to as the aperture, track, or active region where electric fields are produced between fingers **310** to cause an acoustic wave **406** to form at least in this region of the piezoelectric layer **132**.
(52) A physical periodicity of the fingers **310** is referred to as a pitch **408** of the interdigital transducer **304**. The pitch **408** may be indicated in various ways. For example, in certain aspects, the pitch **408** may correspond to a magnitude of a distance between adjacent fingers **310** of the interdigital transducer **304** in the central region **404**. This distance may be defined, for example, as the distance between center points of each of the fingers **310**. The distance may be generally measured between a right (or left) edge of one finger **310** and the right (or left) edge of an adjacent finger **310** when the fingers **310** have uniform widths. In certain aspects, an average of distances between adjacent fingers **310** of the interdigital transducer **304** may be used for the pitch **408**. The frequency at which the piezoelectric layer **132** vibrates is a main-resonance frequency of the electrode structure **130**. The frequency is determined at least in part by the pitch **408** of the interdigital transducer **304** and other properties of the thin-film surface-acoustic-wave filter **126**.
(53) In the three-dimensional perspective view **400-1**, the thin-film surface-acoustic-wave filter **126** is defined by a first (X) axis **410**, a second (Y) axis **412**, and a third (Z) axis **414**. The first axis **410** and the second axis **412** are parallel to a planar surface of the piezoelectric layer **132**, and the second axis **412** is perpendicular to the first axis **410**. The third axis **414** is normal (e.g., perpendicular or orthogonal) to the planar surface of the piezoelectric layer **132**. The busbars **308** of the interdigital transducer **304** are oriented to be parallel to the first axis **410**. The fingers **310** of the interdigital transducer **304** are orientated to be parallel to the second axis **412**. Also, an orientation of the piezoelectric layer **132** causes the acoustic wave **406** to mainly form in a direction of the first axis **410**. As such, the acoustic wave **406** forms in a direction that is substantially perpendicular or orthogonal to the direction of the fingers **310** of the interdigital transducer **304**.
(54) At least one channel **312** is formed or etched within a surface of the piezoelectric layer **132** that faces towards the electrode structure **130**. In the example implementation shown in FIG. **4-1**, the piezoelectric layer **132** has two channels **312**, which are positioned proximate to outer boundaries or edges of the central region **404**. The dielectric material **134** is disposed in the two channels **312**. A surface of the dielectric material **134** that faces towards the electrode structure **130** can be substantially flush with the surface of the piezoelectric layer **132** that also faces towards the electrode structure **130**. In some cases, a surface of the dielectric material **134** that faces away from the electrode structure **130** adheres to the surface of the piezoelectric layer **132** that faces towards the electrode structure **130**. Another example type of surface-acoustic-wave filter **124** is further described with respect to FIG. **4-2**.
(55) FIG. **4-2** illustrates an example implementation of the high-quality temperature-compensated surface-acoustic-wave filter **128** with the dielectric material **134** disposed in the piezoelectric layer **132**. A three-dimensional perspective view **400-3** of the high-quality temperature-compensated surface-acoustic-wave filter **128** is shown at the top of FIG. **4-2**, and a two-dimensional cross-section view **400-4** of the high-quality temperature-compensated surface-acoustic-wave filter **128** is shown at the bottom of FIG. **4-2**.
(56) The high-quality temperature-compensated surface-acoustic-wave filter **128** includes at least one electrode structure **130**, at least one piezoelectric layer **132**, and at least one compensation layer **314**. The compensation layer **314** can provide temperature compensation to enable the high-quality temperature-compensated surface-acoustic-wave filter **128** to achieve a target temperature coefficient of frequency. In example implementations, the compensation layer **314** can be implemented using at least one silicon dioxide layer.
(57) In the depicted configuration shown in the two-dimensional cross-section view **400-4**, the

electrode structure **130** is disposed between the piezoelectric layer **132** and the compensation layer **314**. The piezoelectric layer **132** can form a substrate of the high-quality temperature-compensated surface-acoustic-wave filter **128**.

(58) The electrode structure **130** of the high-quality temperature-compensated surface-acoustic-wave filter **128** can be similar to the electrode structure **130** described above with respect to the thin-film surface-acoustic-wave filter **126** of FIG. **4-1**. Likewise, the piezoelectric layer **132** of the high-quality temperature-compensated surface-acoustic-wave filter **128** can be similar to the piezoelectric layer **132** described above with respect to the thin-film surface-acoustic-wave filter **126** of FIG. **4-1**. The piezoelectric layer **132** of the high-quality temperature-compensated surface-acoustic-wave filter **128**, however, can be thicker than the piezoelectric layer **132** of the thin-film surface-acoustic-wave filter **126** of FIG. **4-1**. Similar to the thin-film surface-acoustic-wave filter **126** of FIG. **4-1**, the high-quality temperature-compensated surface-acoustic-wave filter **128** of FIG. **4-2** can also include the barrier region **402** and the central region **404**.

(59) The high-quality temperature-compensated surface-acoustic-wave filter **128** can have at least one channel **312** formed or etched within a surface of the piezoelectric layer **132** that faces towards the electrode structure **130**. In the example implementation shown in FIG. **4-2**, the piezoelectric layer **132** has two channels, which are positioned proximate to outer boundaries or edges of the central region **404**. The dielectric material **134** is disposed in the two channels **312**. A surface of the dielectric material **134** that faces towards the electrode structure **130** can be substantially flush with the surface of the piezoelectric layer **132** that also faces towards the electrode structure **130**. In some cases, a surface of the dielectric material **134** that faces away from the electrode structure **130** adheres to the surface of the piezoelectric layer **132** that faces towards the electrode structure **130**.

(60) For both the thin-film surface-acoustic-wave filter **126** and the high-quality temperature-compensated surface-acoustic-wave filter **128**, a variety of properties associated with the dielectric material **134** (and the channel **312**) can be tailored to achieve a target level of spurious-mode suppression for a particular range of frequencies. For instance, a density and/or stiffness of the dielectric material **134**, a position of the dielectric material **134** (and channel **312** or multiple channels **312**) within the central region **404**, a length of the dielectric material **134** (and channel **312**) along the first (X) axis **410**, a width of the dielectric material **134** (and channel **312**) along the second (Y) axis **412**, and/or a thickness of the dielectric material **134** (e.g., a depth of the channel **312**) along the third (Z) axis **414** can be designed to attenuate a transversal mode across a range of frequencies by a particular amount. In particular, the width of the dielectric material **134** can be based, at least in part, on the range of frequencies. Additionally, these properties can be tailored to provide a geometry that produces a velocity profile in accordance with a piston mode. For example, the thickness and/or width of the dielectric material **134** can be designed to realize a particular velocity within the trap region **506**.

(61) The thickness of the dielectric material **134** along the third (Z) axis **414** can be between approximately 5% and 100% of twice the pitch **408**. For an example thin-film surface-acoustic-wave filter **126** or an example high-quality temperature-compensated surface-acoustic-wave filter **128**, the thickness of the dielectric material **134** can be approximately 5%, 10%, 25%, 50%, 75%, 80%, or 100% the length of two times the pitch **408**. In cases in which a thickness of the piezoelectric layer **132** is less than twice the pitch **408**, the thickness of the dielectric material **134** can be less than or equal to the thickness of the piezoelectric layer **132**. In general, the term “approximately” can mean that any of the thicknesses can be within +/−10% of a specified value or less (e.g., within +/−5%, +/−3%, or +/−2% of a specified value). Generally, increasing the thickness of the dielectric material **134** decreases the acoustic wave **406**'s velocity within the trap region **506**. Various implementations of the channels **312** and the dielectric material **134** are further described with respect to FIGS. **6-9**. The thin-film surface-acoustic-wave filter **126** of FIG. **4-1** and the high-quality temperature-compensated surface-acoustic-wave filter **128** of FIG. **4-2** can have similar operations, which are generally described below with respect to the surface-acoustic-wave

filter **124**.
(62) During operation, the surface-acoustic-wave filter **124** (e.g., the thin-film surface-acoustic-wave filter **126** of FIG. **4-1** or the high-quality temperature-compensated surface-acoustic-wave filter **128** of FIG. **4-2**) accepts a radio-frequency signal, such as the pre-filter transmit signal **224** or the pre-filter receive signal **230** shown in FIG. **2**. The electrode structure **130** excites an acoustic wave **406** on the piezoelectric layer **132** using the inverse piezoelectric effect. For example, the interdigital transducer **304** in the electrode structure **130** generates an alternating electric field based on the accepted radio-frequency signal. The piezoelectric layer **132** enables the acoustic wave **406** to be formed in response to the alternating electric field generated by the interdigital transducer **304**. In other words, the piezoelectric layer **132** causes, at least partially, the acoustic wave **406** to form responsive to electrical stimulation by one or more interdigital transducers **304**.
(63) The acoustic wave **406** propagates across the piezoelectric layer **132** and interacts with the interdigital transducer **304** or another interdigital transducer within the electrode structure **130** (not shown in FIG. **4-1** or **4-2**). The acoustic wave **406** that propagates can be a standing wave. In some implementations, two reflectors within the electrode structure **130** cause the acoustic wave **406** to be formed as a standing wave across a portion of the piezoelectric layer **132**. In other implementations, the acoustic wave **406** propagates across the piezoelectric layer **132** from the interdigital transducer **304** to another interdigital transducer (not shown).
(64) Using the piezoelectric effect, the electrode structure **130** generates a filtered radio-frequency signal based on the propagated surface acoustic wave **406**. In particular, the piezoelectric layer **132** generates an alternating electric field due to the mechanical stress generated by the propagation of the acoustic wave **406**. The alternating electric field induces an alternating current in the other interdigital transducer or the interdigital transducer **304**. This alternating current forms the filtered radio-frequency signal, which is provided at an output of the surface-acoustic-wave filter **124**. The filtered radio-frequency signal can include the filtered transmit signal **226** or the filtered receive signal **232** of FIG. **2**.
(65) It should be appreciated that while a certain number of fingers are illustrated in FIGS. **4-1** and **4-2**, the number of actual fingers and lengths and width of the fingers and busbars may be different in an actual implementation. Such parameters depend on the particular application and desired filter characteristics. In addition, the thin-film surface-acoustic-wave filter **126** or high-quality temperature-compensated surface-acoustic-wave filter **128** can include multiple interconnected electrode structures each including multiple interdigital transducers **304** to achieve a desired passband (e.g., multiple interconnected resonators or interdigital transducers **304** in series or parallel connections to form a desired filter transfer function).
(66) Although not explicitly shown, the electrode structure **130** can also include two or more reflectors. In an example implementation, the interdigital transducer **304** is arranged between two reflectors (not shown), which reflect the acoustic wave back towards the interdigital transducer **304**. Each reflector within the electrode structure **130** can have two busbars and a grating structure of conductive fingers that each connect to both busbars. In some implementations, the pitch of the reflector can be similar to or the same as the pitch **408** of the interdigital transducer **304** to reflect the acoustic wave **406** in the resonant frequency range.
(67) FIG. **5** illustrates an example electrode structure **130** of the surface-acoustic-wave filter **124** with dielectric material **134** disposed in the piezoelectric layer **132**. The surface-acoustic-wave filter **124** of FIG. **5** can be the thin-film surface-acoustic-wave filter **126** of FIG. **4-1** or the high-quality temperature-compensated surface-acoustic-wave filter **128** of FIG. **4-2**.
(68) As shown in a two-dimensional top-down view **500** of the surface-acoustic-wave filter **124**, the electrode structure **130** of the surface-acoustic-wave filter **124** includes a first busbar **308-1**, a second busbar **308-2**, and fingers **310-1**, **310-2**, **310-3**, and **310-4**. The first busbar **308-1** and the fingers **310-1** and **310-3** form at least a portion of the first comb-shaped structure **306-1**. The fingers **310-1** and **310-3** are connected to the first busbar **308-1** and extend along the second (Y)

axis **412** towards the second busbar **308-2** without connecting to the second busbar **308-2**. The second busbar **308-2** and the fingers **310-2** and **310-4** form at least a portion of the second comb-shaped structure **306-2**. The fingers **310-2** and **310-4** are connected to the second busbar **308-2** and extend along the second (Y) axis **412** towards the first busbar **308-1** without connecting to the first busbar **308-1**.

(69) The surface-acoustic-wave filter **124** includes multiple distinct regions along the second (Y) axis **412**. These regions are defined, at least in part, based on a physical layout of the electrode structure **130**. The regions include a busbar region **502**, the barrier region **402**, and the central region **404**. The busbar region **502** includes the busbars **308-1** and **308-2** and extends along portions of the second (Y) axis **412** that correspond with the widths of the busbars **308-1** and **308-2**. In this case, two busbar regions **502** are shown to be associated with the two busbars **308-1** and **308-2**, respectively.

(70) The barrier region **402** is present between the central region **404** and the busbar region **502**. In particular, the barrier region **402** includes portions of the second (Y) axis **412** that extend between one busbar **308** and the ends of fingers **310** associated with another busbar **308** (e.g., an opposite busbar). For example, a first barrier region **402** exists between the first busbar **308-1** and the ends of fingers **310-2** and **310-4**, which are associated with the second busbar **308-2**. A second barrier region **402** exists between the second busbar **308-2** and ends of fingers **310-1** and **310-3**, which are associated with the first busbar **308-1**.

(71) The central region **404** is defined by the overlap of fingers **310-1** to **310-4** across the first (X) axis **410**. As depicted in the top-down view **500**, the central region **404** includes at least one active track region **504** and at least one trap region **506**. The trap region **506** is present between the barrier region **402** and the active track region **504**. In this way, the trap region **506** exists at the outer boundaries or outer edges of the central region **404**. The channels **312** and the dielectric material **134** are positioned within the trap region **506**. As such, the fingers **310-1** to **310-4** of the electrode structure **130** at least partially cover the dielectric material **134** and the channel **312**. In general, the main or fundamental mode of the surface-acoustic-wave filter **124** is designed to propagate within the active track region **504**. The active track region **504** can also be referred to as a central excitation area.

(72) With the dielectric material **132** disposed in the piezoelectric layer **132**, the structural characteristics of the piezoelectric layer **132** within the trap region **506** can vary from the other regions, including the active track region **504**. This variation causes the acoustic wave **406** (of FIG. **4-1** or **4-2**) to have a lower velocity (e.g., a lower transversal velocity) within the trap region **506** than within the active track region **504**. By disposing the dielectric material **132** within the piezoelectric layer **132**, a velocity profile of the surface-acoustic-wave filter **124** can be designed to reduce (suppress) spurious transversal modes and promote excitation of the main or fundamental wave mode.

(73) The velocity profile indicates velocities (e.g., wave velocities) of each region of the surface-acoustic-wave filter **124**. In an example implementation, the velocity of the acoustic wave **406** is higher within the busbar region **502** and the barrier region **402** in comparison to the central region **404**. In general, the acoustic wave **406** can readily propagate in regions in which the velocity is lower, such as within the central region **404**. The relatively higher velocity within the barrier region **402** and the busbar region **502** effectively forms a barrier, which isolates the central region **404** and reduces leakage (e.g., loss) within the surface-acoustic-wave filter **124**.

(74) Within the central region **404**, the velocity is lower within the trap region **506** in comparison to the active track region **504**. The lower velocity within the trap region **506** can shape the transversal profile (e.g., amplitude) of the fundamental mode. As an example, a difference in velocities between the active track region **504** and the trap region **506** can be on the order of tens or hundreds of meters per second (m/s).

(75) A mode profile of the surface-acoustic-wave filter indicates the amplitude of the fundamental

wave mode across the different regions. In this example implementation, the mode profile has a rectangular or pulse shape, which corresponds to a piston mode in which spurious transversal modes are substantially suppressed (e.g., attenuated). The piston mode is characterized by the amplitude being generally flat (e.g., the same) across the active track region **504** and higher within the active track region **504** in comparison to the busbar and barrier regions **502** and **402**. To realize the piston mode and suppress spurious transversal modes, disposing the dielectric material **134** in the piezoelectric layer **132** increases the mass loading that is applied to the piezoelectric layer **132** within the trap region **506**.

(76) In some implementations, a width of the trap region **506** along the second (Y) axis **412** (e.g., a lateral dimension or transversal dimension of the trap region **506**) can be between approximately 10% to 300% of a wavelength of the acoustic wave **406** along the first (X) axis **410**. For example, the width of the trap region **506** can be between approximately 25% and 100% of the wavelength of the acoustic wave **406**. The width of the trap region **506** along the second (Y) axis **412** can also be specified with respect to the pitch **408** (of FIG. **4-1** or **4-2**) of the electrode structure **130**. In example implementations, the width of the trap region **506** can be between approximately 30% and 200% of the pitch **408** (of FIG. **4-1** or **4-2**) of the electrode structure **130**. For example, the width of the trap region **506** can be approximately 30%, 50%, 100%, or 200% of the pitch **408**. As another example, the width of the trap region **506** can be between approximately 30% and 70% of the pitch **408**. In general, the term “approximately” can mean that any of the widths can be within +/−10% of a specified value or less (e.g., within +/−5%, +/−3%, or +/−2% of a specified value).

(77) For clarity, the regions of the surface-acoustic-wave filter **124** are not necessarily drawn to scale in the top-down view **500**. Further, each region may have some variability that deviates from the illustrated proportions or widths, including variabilities caused by constraints of a given manufacturing technology. For example, the trap region **506** may extend beyond the ends of fingers over which the trap region **506** is disposed. One of ordinary skill in the art can appreciate that these dimensions can vary for other example implementations of the surface-acoustic-wave filter **124**.

(78) In some aspects, the surface-acoustic-wave filter **124** is implemented using additional piston-mode techniques that customize a geometry of the electrode structure **130**. As an example, the fingers **310** within the electrode structure **130** can have varying widths or heights across the length of the fingers **310**. In particular, a finger **310** can have a wider and/or taller profile (e.g., include a hammerhead or a dot, respectively) within the trap region **506** relative to the active track region **504**. The wider and/or taller profile can be realized by applying conducting material, such as metal, and/or a dielectric material. This additional material increases a mass of the fingers **310** within the trap region **506** relative to the active track region **504**.

(79) A portion **508** of the electrode structure **130** is further described with respect to various implementations of the dielectric material **134** in FIGS. **6-9**. As shown in FIG. **5**, the portion **508** includes a “lower” portion (as depicted on the drawing sheet) of the surface-acoustic-wave filter **124**. In particular, the portion **508** includes the trap region **506**, the barrier region **402**, and a portion of the busbar region **502** that is “below” (as depicted on the drawing sheet) the active track region **504**. Although the portion **508** is explicitly depicted in FIGS. **6-9**, an “upper” portion of the surface-acoustic-wave filter **124** (e.g., the trap region **506**, the barrier region **402**, and a portion of the busbar region **502** that is “above” the active track region **504**) can have a similar configuration as the portion **508**. Example configurations of the channel **312** and the dielectric material **134** are further described with respect to FIGS. **6-9**.

(80) FIG. **6** illustrates a first example implementation of the surface-acoustic-wave filter **124** with the dielectric material **134** disposed in the piezoelectric layer **132**. A three-dimensional perspective view **600-1** of the portion **508** of the surface-acoustic-wave filter **124** is shown at the top of FIG. **6**, a two-dimensional cross-section view **600-2** of the surface-acoustic-wave filter **124** is shown at the middle of FIG. **6**, and a two-dimensional top-down view **600-3** of the surface-acoustic-wave filter **124** is shown at the bottom of FIG. **6**.

(81) In the portion **508** illustrated in the three-dimensional perspective view **600-1**, the piezoelectric layer **132** has one channel **312** positioned within the trap region **506** (e.g., within one of the trap regions **506** of FIG. **5**). In particular, the channel **312** is positioned proximate to an end of the finger **310-3** and a beginning of the finger **310-2**. The channel **312** is filled with the dielectric material **134**. As shown in the cross-section view **600-2** and the top-down view **600-3**, the piezoelectric layer **132** can also include another channel **312** within the other trap region **506**. Each channel **312** has a longitudinal axis that is substantially parallel to the first (X) axis (e.g., the direction of acoustic-wave propagation) and substantially perpendicular to the longitudinal axes of the fingers **310**. As shown in the top-down view **600-3**, the fingers **310** of the electrode structure **130** partially cover the dielectric material **134**. As such, some of the dielectric material **134** is exposed.

(82) The implementation shown in FIG. **6** can realize a target amount of spurious-mode suppression for some frequency ranges, such as a low-band frequency range, a mid-band frequency range, or a high-band frequency range. The low-band frequency range can include frequencies below one gigahertz (e.g., long-term evolution (LTE) frequency bands 28 or 71). The mid-band and high-band frequency ranges can include frequencies between one and three gigahertz (e.g., LTE frequency band 41). Multiple channels **312** in the piezoelectric layer **312** within the trap region **506** can also be implemented to achieve target levels of spurious-mode suppression for higher frequency bands (e.g., frequencies between three and six gigahertz (e.g., 5th-generation New Radio (5G NR) frequency bands n77 and n79)) as further described with respect to FIG. **7**.

(83) FIG. **7** illustrates a second example implementation of the surface-acoustic-wave filter **124** with the dielectric material **134** disposed in the piezoelectric layer **132**. A three-dimensional perspective view **700-1** of the portion **508** of the surface-acoustic-wave filter **124** is shown at the top of FIG. **7**, a two-dimensional cross-section view **700-2** of the surface-acoustic-wave filter **124** is shown at the middle of FIG. **7**, and a two-dimensional top-down view **700-3** of the surface-acoustic-wave filter **124** is shown at the bottom of FIG. **7**.

(84) In the portion **508** illustrated in the three-dimensional perspective view **700-1**, the piezoelectric layer **132** has two channels **312-1** and **312-2** positioned within the trap region **506** (e.g., positioned within one of the trap regions **506** of FIG. **5** or positioned proximate to a same boundary of the active track region **504**). The two channels **312-1** and **312-2** can be substantially parallel to each other. The channels **312-1** and **312-2** are filled with the dielectric material **134**.

(85) As shown in the cross-section view **700-2** and the top-down view **700-3**, the piezoelectric layer **132** can also include another set of parallel channels **312** within the other trap region **506**. Each channel **312** has a longitudinal axis that is substantially parallel to the first (X) axis (e.g., the direction of acoustic-wave propagation) and substantially perpendicular to the longitudinal axes of the fingers **310**. As shown in the top-down view **700-3**, the fingers **310** of the electrode structure **130** partially cover the dielectric material **134** disposed in each channel **312**. Some of the dielectric material **134** is exposed.

(86) In general, the piezoelectric layer **132** can be implemented with any quantity of channels **312** within each trap region **506** (e.g., one, two, three, or four channels **312**). Each channel **312** provides another degree of freedom to realize a desired level of spurious-mode suppression performance. With this design, the surface-acoustic-wave filter **124** can achieve the desired level of spurious-mode suppression performance for challenging frequency bands, including ultra-high frequency bands.

(87) In the implementations shown in FIGS. **6** and **7**, the channel **312** is implemented as a continuous channel that spans a track length of the surface-acoustic-wave filter **124** (e.g., spans between outer fingers **310** of the electrode structure **130** along the first (X) axis **410**). In alternative implementations, the channel **312** can instead be implemented using multiple segmented channels that are approximately parallel to the first (X) axis **410**, as further described with respect to FIGS. **8** and **9**.

(88) FIG. **8** illustrates a third example implementation of the surface-acoustic-wave filter **124** with the dielectric material **134** disposed in the piezoelectric layer **132**. A three-dimensional perspective view **800-1** of the portion **508** of the surface-acoustic-wave filter **124** is shown at the top of FIG. **8**, a two-dimensional cross-section view **800-2** of the surface-acoustic-wave filter **124** is shown at the middle of FIG. **8**, and a two-dimensional top-down view **800-3** of the surface-acoustic-wave filter **124** is shown at the bottom of FIG. **8**.
(89) In the portion **508** illustrated in the three-dimensional perspective view **800-1**, the piezoelectric layer **132** has multiple segmented channels **802** positioned within the trap region **506** along the first (X) axis **410** (e.g., positioned within one of the trap regions **506** of FIG. **5** or positioned proximate to a same boundary of the active track region **504**). For example, the piezoelectric layer **132** includes a first segmented channel **802-1** positioned “below” the finger **310-3** and a second segmented channel **802-2** positioned “below” the finger **310-2**. In some cases, the segmented channels **802** are positioned proximate to the ends of the fingers **310**. Optionally, additional segmented channels **802** can be placed proximate to the beginning of the fingers **310**. In general, each segmented channel **802** can be at least partially covered by a corresponding finger **310** of the electrode structure **130**. Adjacent segmented channels **802** are physically separated from each other along the first (X) axis **410**. The longitudinal axes of the segmented channels **802** can be substantially parallel to the first (X) axis (e.g., the direction of acoustic-wave propagation) and substantially perpendicular to the longitudinal axes of the fingers **310**.
(90) As shown in the top-down view **800-3**, the piezoelectric layer **132** can also include other segmented channels **802** within the other trap region **506**. The fingers **310** of the electrode structure **130** partially cover the dielectric material **134**. As such, some of the dielectric material **134** is exposed. In other words, lengths of the segmented channels **802** along the first (X) axis **410** are greater than widths of the fingers **310** along the first (X) axis **410**. In other implementations, the fingers **310** of the electrode structure **130** fully cover the dielectric material **134**, as further described with respect to FIG. **9**.
(91) FIG. **9** illustrates a fourth example implementation of the surface-acoustic-wave filter **124** with the dielectric material **134** disposed in the piezoelectric layer **132**. A three-dimensional perspective view **900-1** of the portion **508** of the surface-acoustic-wave filter **124** is shown at the top of FIG. **9**, a two-dimensional cross-section view **900-2** of the surface-acoustic-wave filter **124** is shown at the middle of FIG. **9**, and a two-dimensional top-down view **900-3** of the surface-acoustic-wave filter **124** is shown at the bottom of FIG. **9**.
(92) In the portion **508** illustrated in the three-dimensional perspective view **900-1**, the piezoelectric layer **132** has multiple segmented channels **802** positioned within the trap region **506** along the first (X) axis **410** (e.g., positioned within one of the trap regions **506** of FIG. **5** or positioned proximate to a same boundary of the active track region **504**). For example, the piezoelectric layer **132** includes a first segmented channel **802-1** positioned “below” the finger **310-3** and a second segmented channel **802-2** positioned “below” the finger **310-2**. In some cases, the segmented channels **802** are positioned proximate to the ends of the fingers **310**. Optionally, additional segmented channels **802** can be placed proximate to the beginning of the fingers **310**. Adjacent segmented channels **802** are physically separated from each other along the first (X) axis **410**. The longitudinal axes of the segmented channels **802** can be substantially parallel to the first (X) axis (e.g., the direction of acoustic-wave propagation) and substantially perpendicular to the longitudinal axes of the fingers **310**.
(93) As shown in the three-dimensional perspective view **900-1** and the top-down view **900-3**, the fingers **310** of the electrode structure **130** fully cover the dielectric material **134**. As such, the dielectric material **134** is not exposed. In other words, lengths of the segmented channels **802** along the first (X) axis **410** are less than widths of the fingers **310** along the first (X) axis **410**.
(94) One of ordinary skill in the art can appreciate the variety of other configurations for which the dielectric material **134** can be partially embedded in the piezoelectric layer **132**. Some

implementations of the piezoelectric layer **132** can have some combination of the designs described in FIGS. **6-9**. For example, the surface-acoustic-wave filter **124** can include a trap region **506** with a first set of segmented channels **802** as described in FIG. **8** or **9** and a second set of segmented channels **802** that is substantially parallel to the first set of segmented channels along the first (X) axis **410** and offset from the first set of segmented channels along the second (Y) axis **412**. Explained another way, the first set of segmented channels **802** and the second set of segmented channels **802** can be positioned proximate to a same boundary of the active track region **504**. As another example, the surface-acoustic-wave filter **124** can include more than two parallel channels **312** within a trap region **506** (e.g., positioned within one of the trap regions **506** of FIG. **5** or positioned proximate to a same boundary of the active track region **504**), such as three or four channels **312**.

(95) Although the segmented channels **802** of FIGS. **8** and **9** are shown to be aligned along a same axis that is substantially parallel to the first (X) axis **410**, other implementations of the segmented channels **802** can apply an offset along the second (Y) axis **412** for different sets of the segmented channels **802**. For example, a first set of the segmented channels **802** can be aligned along a first axis that is substantially parallel to the first (X) axis **410** and intersects the second (Y) axis **412** at a first point. A second set of the segmented channels **802** can be aligned along a second axis that is substantially parallel to the first (X) axis **410** and intersects the second (Y) axis **412** at a second point, which differs from the first point.

(96) Although the surface-acoustic-wave filters **124** of FIGS. **6-9** are illustrated as thin-film surface-acoustic-wave filters **126** with a substrate layer **302**, other implementations of the surface-acoustic-wave filter **124** are also possible. For example, the surface-acoustic-wave filters **124** of FIGS. **6-9** can be implemented as high-quality temperature-compensated surface-acoustic-wave filters **128** with a piezoelectric layer **132** that functions as the substrate layer **302**.

(97) FIG. **10** illustrates example frequency responses for implementations of a surface-acoustic-wave filter **124** that does and does not include the dielectric material **134** disposed in the piezoelectric layer **132**. A graph **1000** depicts a frequency response **1002** of another surface-acoustic-wave filter that does not include the dielectric material **134** disposed in the piezoelectric layer **132**. Instead of the dielectric material **134** disposed in the piezoelectric layer **132**, the other surface-acoustic-wave filter includes a dielectric bar that is disposed across fingers of the electrode structure. The dielectric bar enables the other surface-acoustic-wave filter to operate in the piston mode.

(98) The graph **1000** also depicts a frequency response **1004** of the surface-acoustic-wave filter **124**, which includes the dielectric material **134** disposed in the piezoelectric layer **132**. Due to the existence of the dielectric material **134**, the frequency response **1004** of the surface-acoustic-wave filter **124** can attenuate spurious modes by a greater amount than the other surface-acoustic-wave filter, as shown at **1006**. In some cases, this attenuation can be on the order of a few decibels (dB) or more (e.g., approximately 2, 3, 5, 8, or 10 dB). In general, the term "approximately" can mean that the attenuation can be within +/−10% of a specified value or less (e.g., within +/−5%, +/−3%, or +/−2% of a specified value).

(99) FIG. **11** is a flow diagram illustrating an example process **1100** for manufacturing a surface-acoustic-wave filter **124** with dielectric material disposed in a piezoelectric layer. The process **1100** is described in the form of a set of blocks **1102-1108** that specify operations that can be performed. However, operations are not necessarily limited to the order shown in FIG. **11** or described herein, for the operations may be implemented in alternative orders or in fully or partially overlapping manners. Also, more, fewer, and/or different operations may be implemented to perform the process **1100**, or an alternative process. Operations represented by the illustrated blocks of the process **1100** may be performed to manufacture the surface-acoustic-wave filter **124** (e.g., of FIGS. **1** to **3**). More specifically, the operations of the process **1100** may be performed, at least in part, to dispose dielectric material **134** in a piezoelectric layer **132** of a surface-acoustic-wave filter **124**

(e.g., of FIGS. **6-9**).
(100) At **1102**, a piezoelectric layer is provided. For example, the manufacturing process provides the piezoelectric layer **132**. The piezoelectric layer **132** can be implemented using various materials, such as lithium niobate, lithium tantalate, quartz, aluminium nitride, aluminium scandium nitride, or some combination thereof.
(101) At **1104**, at least one channel is etched across a surface of the piezoelectric layer. For example, the manufacturing process etches at least one channel **312** across a surface of the piezoelectric layer **132**. The channel **312** can correspond to a continuous channel that spans the track length of the surface-acoustic-wave filter **124** proximate to a boundary of the active track region **504**, as shown in FIG. **6** or **7**. Alternatively, the channel **312** can correspond to one of multiple segmented channels **802** that are positioned along the track length proximate to a boundary of the active track region **504**, as shown in FIG. **8** or **9**.
(102) In some implementations, multiple channels **312** are etched across the surface of the piezoelectric layer **132**. The multiple channels **312** can be positioned in a same trap region **506** (e.g., proximate to a same boundary of the active track region **504**) or in different trap regions **506** (e.g., proximate to different boundaries of the active track region **504**). Properties of the channel **312** (e.g., position, length, width, or depth) can be tailored to realize a particular level of spurious-mode suppression. In some implementations, the piezoelectric layer **132** is formed to have a channel **312** therein.
(103) At **1106**, dielectric material is deposited within the at least one channel. For example, the manufacturing process deposits the dielectric material **134** within the at least one channel **312**. In some cases, the manufacturing process fills the at least one channel **312** with the dielectric material **134** such that a surface of the dielectric material **134** is substantially flush with the surface of the piezoelectric layer **132**. The dielectric material **134** can include tantalum oxide, silicon oxide, silicon dioxide, silicon nitride, hafnium oxide, aluminum oxide, tungsten trioxide, niobium pentoxide, or some combination or doped version thereof. The density and/or stiffness of the dielectric material **134** can be tailored to realize a particular level of spurious-mode suppression.
(104) At **1108**, an electrode structure that at least partially covers the dielectric material is provided. For example, the manufacturing process provides the electrode structure **130**, which at least partially covers the dielectric material **134**. The electrode structure **130** can partially cover the dielectric material **134**, as shown in FIGS. **6-8**, or fully cover the dielectric material **134**, as shown in FIG. **9**. In some implementations, the electrode structure **130** is in physical contact with the dielectric material **134** or separated by another dielectric layer.
(105) Some aspects are described below.
(106) Aspect 1: An apparatus comprising: a surface-acoustic-wave filter comprising: a piezoelectric layer having at least one channel; an electrode structure; and dielectric material disposed in the at least one channel of the piezoelectric layer, the dielectric material at least partially covered by the electrode structure.
(107) Aspect 2: The apparatus of aspect 1, wherein: the electrode structure comprises: a first busbar and a second busbar; a first set of fingers extending from the first busbar toward the second busbar; and a second set of fingers extending from the second busbar toward the first busbar; and the at least one channel comprises: a first channel that is proximate to ends of fingers associated with the first set of fingers; and a second channel that is proximate to ends of fingers associated with the second set of fingers.
(108) Aspect 3: The apparatus of aspect 1 or 2, wherein: the electrode structure comprises at least one finger; and the at least one channel has a longitudinal axis that is substantially orthogonal to a longitudinal axis of the at least one finger.
(109) Aspect 4: The apparatus of aspect 3, wherein: the at least one channel comprises one channel that is proximate to a boundary of an active track region of the surface-acoustic-wave filter; and the dielectric material disposed in the one channel is at least partially covered by the at least one finger.

(110) Aspect 5: The apparatus of any previous aspect, wherein: the electrode structure comprises multiple fingers; and the dielectric material is at least partially covered by the multiple fingers.
(111) Aspect 6: The apparatus of aspect 5, wherein: the at least one channel comprises two channels that are approximately parallel to each other and are proximate to a boundary of an active track region of the surface-acoustic-wave filter; the dielectric material disposed in a first channel of the two channels is at least partially covered by the multiple fingers; and the dielectric material disposed in a second channel of the two channels is at least partially covered by the multiple fingers.
(112) Aspect 7: The apparatus of aspect 5, wherein: the at least one channel comprises multiple channels; and the dielectric material disposed in each channel of the multiple channels is at least partially covered by a corresponding finger of the multiple fingers.
(113) Aspect 8: The apparatus of aspect 7, wherein the multiple channels are arranged along an axis associated with a direction of acoustic-wave propagation.
(114) Aspect 9: The apparatus of aspect 7 or 8, wherein longitudinal axes of the multiple channels are substantially orthogonal to longitudinal axes of the multiple fingers.
(115) Aspect 10: The apparatus of any one of aspects 7-9, wherein the dielectric material disposed in each channel of the multiple channels is fully covered by the corresponding finger of the multiple fingers.
(116) Aspect 11: The apparatus of any previous aspect, wherein a surface of the dielectric material that faces towards the electrode structure is substantially flush with a surface of the piezoelectric layer that faces towards the electrode structure.
(117) Aspect 12: The apparatus of any previous aspect, wherein: the electrode structure has a pitch; and a transversal axis of the dielectric material has a width between approximately 30% and 200% of the pitch.
(118) Aspect 13: The apparatus of any previous aspect, wherein a thickness of the dielectric material is between approximately 5% and 100% of a length of two times a pitch of the electrode structure.
(119) Aspect 14: The apparatus of aspect 13, wherein the thickness of the dielectric material is approximately 50% of the length of two times the pitch of the electrode structure.
(120) Aspect 15: The apparatus of aspect 13, wherein the thickness of the dielectric material is between approximately 5% and 20% the length of two times the pitch of the electrode structure.
(121) Aspect 16: The apparatus of any previous aspect, wherein the dielectric material comprises at least one of the following: tantalum oxide (Ta.sub.2O.sub.5); silicon oxide (SiO), silicon dioxide (SiO.sub.2); silicon nitride (Si.sub.3N.sub.4); hafnium oxide (HfO.sub.2); aluminium oxide (Al.sub.2O.sub.3); tungsten trioxide (WO.sub.3); or niobium pentoxide (Nb.sub.2O.sub.5).
(122) Aspect 17: The apparatus of any previous aspect, wherein: the surface-acoustic-wave filter comprises multiple cascaded resonators; and a resonator of the multiple cascaded resonators comprises the piezoelectric layer, the electrode structure, and the dielectric material.
(123) Aspect 18: The apparatus of any previous aspect, wherein: the surface-acoustic-wave filter comprises a substrate layer; the substrate layer comprises a compensation layer; and the piezoelectric layer is disposed between the electrode structure and the compensation layer of the substrate layer.
(124) Aspect 19: The apparatus of any previous aspect, wherein the surface-acoustic-wave filter comprises a thin-film surface-acoustic-wave filter.
(125) Aspect 20: The apparatus of any one of aspects 1-18, wherein the surface-acoustic-wave filter comprises a high-quality temperature-compensated surface-acoustic-wave filter.
(126) Aspect 21: An apparatus comprising: a surface-acoustic-wave filter configured to generate a filtered signal from a radio-frequency signal, the surface-acoustic-wave filter comprising: means for converting the radio-frequency signal to an acoustic wave and converting a propagated acoustic wave into the filtered signal; means for propagating the acoustic wave across a planar surface to

produce the propagated acoustic wave; and means for operating in a piston mode, the means for operating disposed in the means for propagating and at least partially covered by the means for converting.
(127) Aspect 22: The apparatus of aspect 21, wherein the means for operating comprises means for attenuating transversal modes.
(128) Aspect 23: The apparatus of aspect 21 or 22, wherein the means for operating is present within a trap region of the surface-acoustic-wave filter.
(129) Aspect 24: A method of manufacturing a surface-acoustic-wave filter, the method comprising: providing a piezoelectric layer; etching at least one channel across a surface of the piezoelectric layer; depositing dielectric material within the at least one channel; and providing an electrode structure that at least partially covers the dielectric material.
(130) Aspect 25: The method of aspect 24, wherein the depositing of the dielectric material comprises filling the at least one channel with the dielectric material such that a surface of the dielectric material that faces the electrode structure is substantially flush with the surface of the piezoelectric layer.
(131) Aspect 26: An apparatus comprising: a piezoelectric layer configured to propagate an acoustic wave along a first axis, the piezoelectric layer having at least one recessed area with a longitudinal axis that is substantially parallel to the first axis; and dielectric material positioned in the at least one recessed area, the dielectric material having at least one surface exposed.
(132) Aspect 27: The apparatus of aspect 26, further comprising: an electrode structure that is in contact with the piezoelectric layer and the dielectric material.
(133) Aspect 28: The apparatus of aspect 27, further comprising: a surface-acoustic-wave filter comprising the piezoelectric layer, the dielectric material, and the electrode structure.
(134) Aspect 29: The apparatus of aspect 28, wherein the dielectric material is present within a trap region of the surface-acoustic-wave filter.
(135) Aspect 30: The apparatus of aspect 29, wherein: the surface-acoustic-wave filter is configured to generate the acoustic wave such that the acoustic wave has a first velocity within an active track region of the surface-acoustic-wave filter and a second velocity within the trap region; and the second velocity is lower than the first velocity.
(136) Unless context dictates otherwise, use herein of the word “or” may be considered use of an “inclusive or,” or a term that permits inclusion or application of one or more items that are linked by the word “or” (e.g., a phrase “A or B” may be interpreted as permitting just “A,” as permitting just “B,” or as permitting both “A” and “B”). As used herein, a phrase referring to “at least one of” a list of items refers to any combination of those items, including single members. As an example, “at least one of: a, b, or c” is intended to cover: a, b, c, a-b, a-c, b-c, and a-b-c, as well as any combination with multiples of the same element (e.g., a-a, a-a-a, a-a-b, a-a-c, a-b-b, a-c-c, b-b, b-b-b, b-b-c, c-c, and c-c-c or any other ordering of a, b, and c). Further, items represented in the accompanying figures and terms discussed herein may be indicative of one or more items or terms, and thus reference may be made interchangeably to single or plural forms of the items and terms in this written description. Finally, although subject matter has been described in language specific to structural features or methodological operations, it is to be understood that the subject matter defined in the appended claims is not necessarily limited to the specific features or operations described above, including not necessarily being limited to the organizations in which features are arranged or the orders in which operations are performed.

Claims

1. An apparatus comprising: a surface-acoustic-wave filter comprising: a piezoelectric layer having a plurality of segmented channels; an electrode structure comprising multiple fingers; and dielectric material disposed in each segmented channel of the plurality of segmented channels of the

piezoelectric layer, the dielectric material in each segmented channel of the plurality of segmented channels at least partially covered by a corresponding finger of the multiple fingers of the electrode structure, wherein each segmented channel of the plurality of segmented channels extends laterally beyond the corresponding finger of the multiple fingers.
2. The apparatus of claim 1, wherein: the electrode structure comprises: a first busbar and a second busbar; a first set of fingers of the multiple fingers extending from the first busbar toward the second busbar; and a second set of fingers of the multiple fingers extending from the second busbar toward the first busbar, wherein the corresponding finger is part of the first set of fingers or the second set of fingers; and the plurality of segmented channels comprises: a first plurality of segmented channels that is proximate to ends of fingers associated with the first set of fingers; and a second plurality of segmented channels that is proximate to ends of fingers associated with the second set of fingers.
3. The apparatus of claim 1, wherein the plurality of segmented channels have a longitudinal axis that is substantially orthogonal to a longitudinal axis of the corresponding finger.
4. The apparatus of claim 3, wherein: the plurality of segmented channels comprises one channel that is proximate to a boundary of an active track region of the surface-acoustic-wave filter; and the dielectric material disposed in the one channel is at least partially covered by the corresponding finger.
5. The apparatus of claim 1, wherein the dielectric material in each segmented channel of the plurality of segmented channels is at least partially covered by an end of the corresponding finger of the multiple fingers.
6. The apparatus of claim 5, wherein: the plurality of segmented channels comprises a first plurality of segmented channels and a second plurality of segmented channels, wherein the first plurality of segmented channels is approximately parallel to the second plurality of segmented channels and is proximate to a boundary of an active track region of the surface-acoustic-wave filter; the dielectric material disposed in the first plurality of segmented channels is at least partially covered by the multiple fingers; and the dielectric material disposed in the second plurality of segmented channels is at least partially covered by the multiple fingers.
7. The apparatus of claim 1, wherein the surface-acoustic-wave filter comprises a temperature-compensated surface-acoustic-wave filter.
8. The apparatus of claim 1, wherein the plurality of segmented channels is arranged along an axis associated with a direction of acoustic-wave propagation.
9. The apparatus of claim 1, wherein longitudinal axes of the plurality of segmented channels are substantially orthogonal to longitudinal axes of the multiple fingers.
10. The apparatus of claim 1, wherein the dielectric material disposed in each segmented channel of the plurality of segmented channels is fully covered by the corresponding finger of the multiple fingers.
11. The apparatus of claim 1, wherein a surface of the dielectric material that faces towards the electrode structure is substantially flush with a surface of the piezoelectric layer that faces towards the electrode structure.
12. The apparatus of claim 1, wherein: the electrode structure has a pitch; and a transversal axis of the dielectric material has a width between approximately 30% and 200% of the pitch.
13. The apparatus of claim 1, wherein a thickness of the dielectric material is between approximately 5% and 100% of a length of two times a pitch of the electrode structure.
14. The apparatus of claim 13, wherein the thickness of the dielectric material is approximately 50% of the length of two times the pitch of the electrode structure.
15. The apparatus of claim 13, wherein the thickness of the dielectric material is between approximately 5% and 20% of the length of two times the pitch of the electrode structure.
16. The apparatus of claim 1, wherein the dielectric material comprises at least one of the following: tantalum oxide (Ta2O5); silicon oxide (SiO); silicon dioxide (SiO2); silicon nitride

(SiN); hafnium oxide (HfO2); aluminium oxide (Al2O3); tungsten trioxide (WO3); or niobium pentoxide (Nb2O5).

17. The apparatus of claim 1, wherein: the surface-acoustic-wave filter comprises multiple cascaded resonators; and a resonator of the multiple cascaded resonators comprises the piezoelectric layer, the electrode structure, and the dielectric material.

18. The apparatus of claim 1, wherein: the surface-acoustic-wave filter comprises a substrate layer; the substrate layer comprises a compensation layer; and the piezoelectric layer is disposed between the electrode structure and the compensation layer of the substrate layer.

19. The apparatus of claim 1, wherein the surface-acoustic-wave filter comprises a thin-film surface-acoustic-wave filter (TFSAW).

20. An apparatus comprising: a piezoelectric layer configured to propagate an acoustic wave along a first axis, the piezoelectric layer having a plurality of segmented recessed areas with a longitudinal axis that is substantially parallel to the first axis; an electrode structure comprising multiple fingers; and dielectric material positioned in the plurality of segmented recessed areas, the dielectric material having at least one surface exposed, the dielectric material in each segmented recessed area at least partially covered by a corresponding finger of the multiple fingers of the electrode structure, wherein each segmented recessed area of the plurality of segmented recessed areas extends laterally beyond the corresponding finger of the multiple fingers.

21. The apparatus of claim 20, the electrode structure is in contact with the piezoelectric layer and the dielectric material.

22. The apparatus of claim 21, further comprising: a surface-acoustic-wave filter comprising the piezoelectric layer, the dielectric material, and the electrode structure.

23. The apparatus of claim 22, wherein the dielectric material is present within a trap region of the surface-acoustic-wave filter.

24. The apparatus of claim 23, wherein: the surface-acoustic-wave filter is configured to generate the acoustic wave such that the acoustic wave has a first velocity within an active track region of the surface-acoustic-wave filter and a second velocity within the trap region; and the second velocity is lower than the first velocity.